IP Addressing Services

Domain Name System

There are other application layer-specific protocols that were designed to make it easier to obtain addresses for network devices. These services are essential because it would be very time consuming to remember IP addresses instead of URLs or manually configure all of the devices in a medium to large network. The first topic in this module gave you an overview of these protocols. This topic goes into more detail about the IP addressing services, DNS and DHCP.

In data networks, devices are labeled with numeric IP addresses to send and receive data over networks. Domain names were created to convert the numeric address into a simple, recognizable name.

On the Internet, fully-qualified domain names (FQDNs), such as http://www.cisco.com, are much easier for people to remember than 198.133.219.25, which is the actual numeric address for this server. If Cisco decides to change the numeric address of www.cisco.com, it is transparent to the user because the domain name remains the same. The new address is simply linked to the existing domain name and connectivity is maintained.

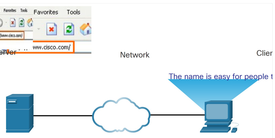
The DNS protocol defines an automated service that matches resource names with the required numeric network address. It includes the format for queries, responses, and data. The DNS protocol communicates using a single format called a message. This message format is used for all types of client queries and server responses, error messages, and the transfer of resource record information between servers.

Select each tab for more information.

Step 1 | Step 2 | Step 3 | Step 4 | Step 5

Step 1

The user types an FQDN into a browser application Address field.



DNS Message Format

The DNS server stores different types of resource records that are used to resolve names. These records contain the name, address, and type of record. Some of these record types are as follows:

- **A** – An end device IPv4 address
- **NS** – An authoritative name server
- **AAAA** – An end device IPv6 address (pronounced quad-A)
- **MX** – A mail exchange record

When a client makes a query, the server DNS process first looks at its own records to resolve the name. If it is unable to resolve the name by using its stored records, it contacts other servers to resolve the name. After a match is found and returned to the original requesting server, the server temporarily stores the numbered address in the event that the same name is requested again.

The DNS client service on Windows PCs also stores previously resolved names in memory. The **ipconfig /displaydns** command displays all of the cached DNS entries.

As shown in the table, DNS uses the same message format between servers, consisting of a question, answer, authority, and additional information for all types of client queries and server responses, error messages, and transfer of resource record information.

DNS message section	Description
Question	The question for the name server
Answer	Resource Records answering the question
Authority	Resource Records pointing toward an authority
Additional	Resource Records holding additional information

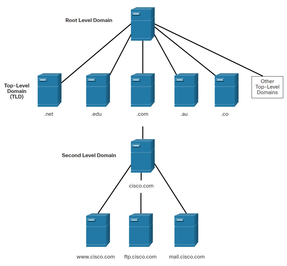
DNS Hierarchy

The DNS protocol uses a hierarchical system to create a database to provide name resolution, as shown in the figure. DNS uses domain names to form the hierarchy.

The naming structure is broken down into small, manageable zones. Each DNS server maintains a specific database file and is only responsible for managing name-to-IP mappings for that small portion of the entire DNS structure. When a DNS server receives a request for a name translation that is not within its DNS zone, the DNS server forwards the request to another DNS server within the proper zone for translation. DNS is scalable because hostname resolution is spread across multiple servers.

The different top-level domains represent either the type of organization or the country of origin. Examples of top-level domains are the following:

- **.com** – a business or industry
- **.org** – a non-profit organization
- **.au** – Australia
- **.co** – Colombia



The nslookup Command

When configuring a network device, one or more DNS Server addresses are provided that the DNS client can use for name resolution. Usually the ISP provides the addresses to use for the DNS servers. When a user application requests to connect to a remote device by name, the requesting DNS client queries the name server to resolve the name to a numeric address.

Computer operating systems also have a utility called Nslookup that allows the user to manually query the name servers to resolve a given host name. This utility can also be used to troubleshoot name resolution issues and to verify the current status of the name servers.

In this figure, when the **nslookup** command is issued, the default DNS server configured for your host is displayed. The name of a host or domain can be entered at the **nslookup** prompt. The Nslookup utility has many options available for extensive testing and verification of the DNS process.

```
C:\Users> nslookup
Default Server:  dns-c1.cisco.com
Address:  171.70.168.183
> www.cisco.com
Server:  dns-c1.cisco.com
Address:  171.70.168.183
Non-authoritative answer:
Name:    GRP-C010-LV3-B
         173.37.145.84
Aliases: www.cisco.com
> cisco.netacad.net
Server:  dns-c1.cisco.com
Address:  171.70.168.183
Name:    cisco.netacad.net
Address:  72.163.5.201
>
```

Syntax Checker – The nslookup Command

Practice entering the nslookup command in both Windows and Linux

From the Windows command prompt, enter the **nslookup** command to begin a manual query of the name servers.

```
C:\> nslookup
Default Server: unknown
Address: 10.10.10.1

The output lists the name and IP address of the DNS server configured in the client. Note that the DNS server address can be manually configured, or
dynamically learned, through DHCP. You are now in nslookup mode. Enter the domain name www.cisco.com.

>
```

Reset | Show Me | Show All

Dynamic Host Configuration Protocol

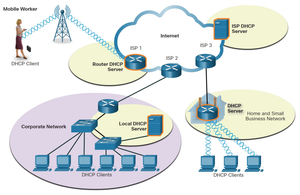
The Dynamic Host Configuration Protocol (DHCP) for IPv4 service automates the assignment of IPv4 addresses, subnet masks, gateways, and other IPv4 networking parameters. This is referred to as dynamic addressing. The alternative to dynamic addressing is static addressing. When using static addressing, the network administrator manually enters IP address information on hosts.

When a host connects to the network, the DHCP server is contacted, and an address is requested. The DHCP server chooses an address from a configured range of addresses called a pool and assigns (leases) it to the host.

On larger networks, or where the user population changes frequently, DHCP is preferred for address assignment. New users may arrive and need connections; others may have new computers that must be connected. Rather than use static addressing for each connection, it is more efficient to have IPv4 addresses assigned automatically using DHCP.

DHCP can allocate IP addresses for a configurable period of time, called a lease period. The lease period is an important DHCP setting. When the lease period expires or the DHCP server gets a DHCPRELEASE message the address is returned to the DHCP pool for reuse. Users can freely move from location to location and easily re-establish network connections through DHCP.

As the figure shows, various types of devices can be DHCP servers. The DHCP server in most medium-to-large networks is usually a local, dedicated PC-based server. With home networks, the DHCP server is usually located on the local router that connects the home network to the ISP.

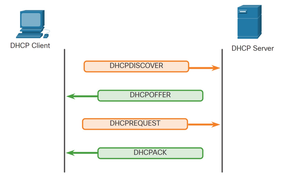


Many networks use both DHCP and static addressing. DHCP is used for general purpose hosts, such as end user devices. Static addressing is used for network devices, such as gateway routers, switches, servers, and printers.

DHCP for IPv6 (DHCPv6) provides similar services for IPv6 clients. One important difference is that DHCPv6 does not provide a default gateway address. This can only be obtained dynamically from the Router Advertisement message of the router.

DHCP Operation

As shown in the figure, when an IPv4, DHCP-configured device boots up or connects to the network, the client broadcasts a DHCP discover (DHCPDISCOVER) message to identify any available DHCP servers on the network. A DHCP server replies with a DHCP offer (DHCPOFFER) message, which offers a lease to the client. The offer message contains the IPv4 address and subnet mask to be assigned, the IPv4 address of the DHCP server, and the IPv4 address of the default gateway. The lease offer also includes the duration of the lease.



The client may receive multiple DHCPOFFER messages if there is more than one DHCP server on the local network. Therefore, it must choose between them, and sends a DHCP request (DHCPREQUEST) message that identifies the explicit server and lease offer that the client is accepting. A client may also choose to request an address that it had previously been allocated by the server.

Assuming that the IPv4 address requested by the client, or offered by the server, is still available, the server returns a DHCP acknowledgment (DHCPACK) message that acknowledges to the client that the lease has been finalized. If the offer is no longer valid, then the selected server responds with a DHCP negative acknowledgment (DHCPNAK) message. If a DHCPNAK message is returned, then the selection process must begin again with a new DHCPDISCOVER message being transmitted. After the client has the lease, it must be renewed prior to the lease expiration through another DHCPREQUEST message.

The DHCP server ensures that all IP addresses are unique (the same IP address cannot be assigned to two different network devices simultaneously). Most ISPs use DHCP to allocate addresses to their customers.

DHCPv6 has a set of messages that is similar to those for DHCPv4. The DHCPv6 messages are SOLICIT, ADVERTISE, INFORMATION REQUEST, and REPLY.

Lab – Observe DNS Resolution

In this lab, you will complete the following objectives:

- Part 1: Observe the DNS Conversion of a URL to an IP Address
- Part 2: Observe DNS Lookup Using the **nslookup** Command on a Web Site
- Part 3: Observe DNS Lookup Using the **nslookup** Command on Mail Servers

Observe DNS Resolution

Check Your Understanding – IP Addressing Services

Check your understanding of DNS services by choosing the correct answer to the following questions.

1. Which of the following DNS record types is used to resolve IPv6 addresses?
   - A
   - NS
   - AAAA
   - MX
2. True or false? A DNS server that receives a request for a name resolution that is not within its DNS zone will send a failure message to the requesting client.
   - True
   - False
3. Which of the following is displayed by the nslookup utility?
   - the configured default DNS server
   - the IP address of the end device
   - all cached DNS entries
4. Which of the following DNS resource record types resolves authoritative name servers?
   - NS
   - A
   - MX
   - AAAA

Check | Show Me | Reset